Name : Koshal Jha

Roll No. : 21/441

Branch : Information Technology

Batch : G - 2

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Assignment-1

Q.1) What is learning? Explain different types of learning.

Ans.) learning is a process that improves the knowledge of an AI program by making observations about its environment. types of learning are:-

(•) Supervised learning -

→ Supervised learning is the process of making an algorithm to learn to map an input to a particular output. This is achieved using the labelled datasets that you have collected

(•) Unsupervised learning -

Unsupervised learning is a machine learning technique in which the users donot need to supervise the model. It allows the model to work on its own to discover patterns and information that was previously undetected. It mainly deals with the unlabeled data.

(•) Reinforcement learning -

Reinforcement learning method that is concerned with how software agents should take actions in an environment. This neural network learning method helps you to learn how to attain a complex objective or maximize a specific dimension over many steps.

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Q2.) What is data analytics. Explain various steps involved in it.

Ans.) It is defined as a process of cleaning, transforming, and modeling data to discover useful information for business decision-making.

The various steps involved in data analytics :-

Step 1:- Define questions & goals.

The first step in data analysis is to clearly define your questions and goals. Similar to creating a hypothesis before an experiment, you should be asking a targeted question before searching the data for an answer.

Step 2 :- Collect data.

Before you can start analyzing, there needs to be data available for use. Data can include sales records, customer demographics, lead tracking, net promoter scores, and more. When using a business intelligence tool, it is important to make sure that all of the data is accessible and the proper connections are set between your data warehouse and your BI tool of choice.

Step 3 :- Data Wrangling.

Now that you have all of your data in one place, it is important to clean the data before ~~begning~~ beginning the analysis portion of this process. A large part of the ~~clearing~~ cleansing process includes making sure that the data is in a usable format. This entails searching for outliers, dealing with null value and looking for data that may have been incorrectly input. often this can be a lengthy and arduous process

Step4 :- Determine analysis.
once the relevant data is ~~an~~ available and cleansed, it is time to analyze. Choosing a method of analysis will heavily depend on the question or goals defined earlier and the type of analysis needed. ~~Diago~~ Diagnostic analysis uses data to search for the cause of the cause of, and a solution, to an existing problem. Descriptive analysis is a way to describe the data by summarizing key sections.

Step5 :- Interpret results.
After analyzing your data, it is important to interpret the results. Simply put, what are you learning from the results of your analysis? One way to interpret the results is by creating data visualizations. These can be done with coding libaries such as plotly or matplotlib. But many BI tools are also avaiable and will enable users to visualize data in many different ways.

Q3:) Explain DFS and BFS with examples.

Ans:) Depth first search (D.F.S) :-
Depth first search (DFS) algorithm traverses a graph in a depthward motion and uses a stack to remember to get the next vertex to start a search, when a dead end occurs in any iteration.
As in the example given above, DFS algorithm traverses from S to A to D to G to E to B first, then ~~the~~ to F and lastly to C.

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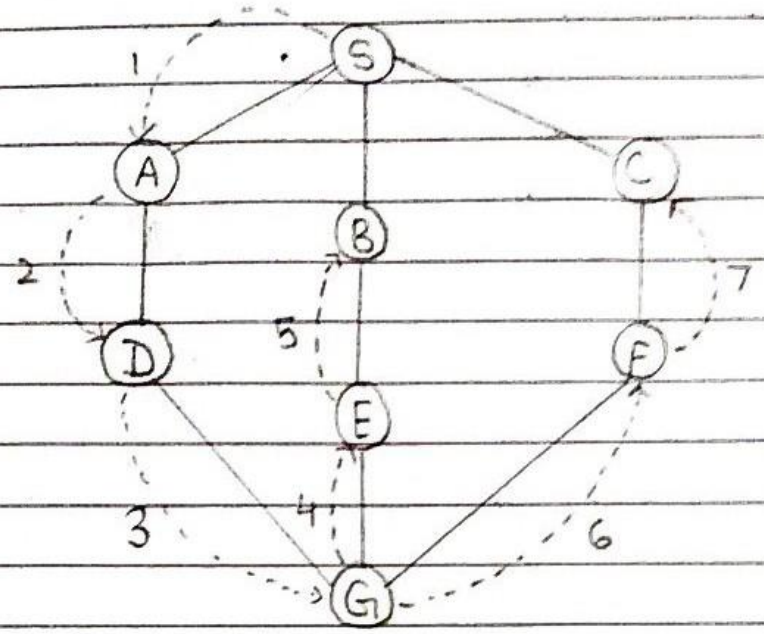
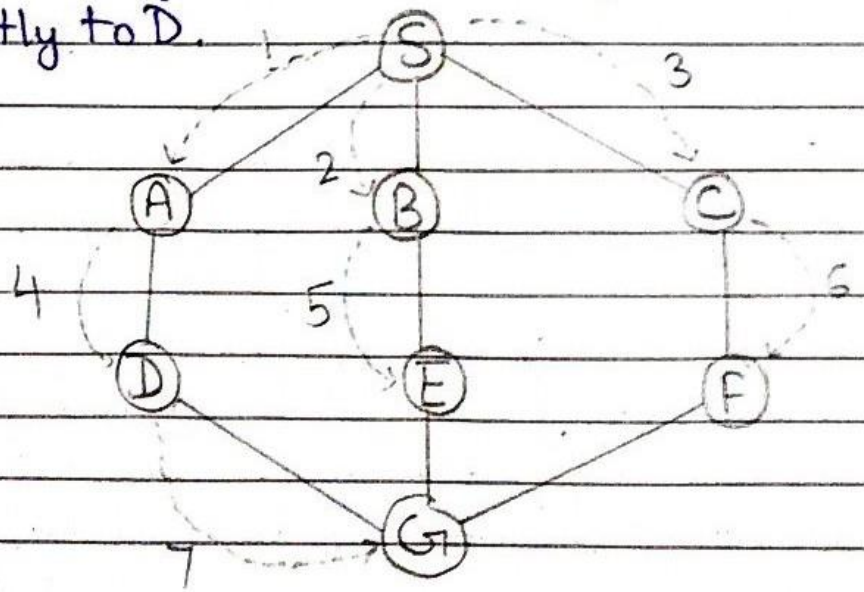


Breadth first search (B.F.S) :-

Breadth first search (BFS) algorithm traverse a graph in a breadthward motion and uses a queue to remember to get the next vertex to start a search, when a dead end occurs in any iteration.

As in the example given below, BFS algorithm traverses from A to B to E to F first then to C and G lastly to D.



Q4.) what is turning test? Explain.

Ans.) The turning test is a deceptively simple method of determining whether a machine can demostrate human intelligence. if a machine can engage in a conservation with a human without being detected as a machine, it has demostrated human intelligence. The turning test was proposed in a paper published in 1950 by mathematician and computing pioneer Alan Turning. It has become a fundamental motivator in the theory and development of artifical intelligence. It measures the intelligence of a test subject to determine whether a machine can demostrate intelligence, according to the test, a computer program can think if its reponses can fool a human into believing it, too, is human. Turning concludes that if the judge can't tell the difference, the computer has succeeded in demonstrating human intelligence.

Q5.) What is graph and types of graphs?

Ans.) A graph ~~structure~~ data structure is made up of a finite and potentially mutable set of vertices (also known as nodes or points), as well as a set of unordered pairs for an undirected graph or a set of ordered pairs for a directed graphs. These pairs are recognized as edges, links, or lines in a directed graph but are also known as arrows or arcs. The vertices could be internal graph elements

represented by integer indices or references.
The types of graphs are :-

→ Null Graph :- The term "null graph" refers to a graph with an empty edge set.

→ Trivial Graph :- The trivial graph is the smallest possible graph that can be created with the least number of vertices that is only one vertex only.

→ Non-Directed Graph :- A graph is called a non-directed graph if all the edges present between any graph nodes are non-directed so it mean we cannot determine from where node is starting and at which node it is ending.

→ Directed graph :- A graph is called directed graph if all the edges present between any vertices or nodes of the graph are directed or have a defined direction.

→ Connected Graph :- For a graph to be labelled as a connected graph, there must be at least a single path between every pair of the graph vertices.

→ Disconnected Graph :- A graph is said to be disconnected graph where there does not exist any path between at least one pair of vertices.